



GOSFORD HIGH SCHOOL

2012

HSC ASSESSMENT TASK 1

EXTENSION 1 MATHEMATICS

General Instructions:

Attempt all Questions 1-3

- Working time: 60 minutes
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question should be started on a separate page.
- All necessary working should be shown in every question.

Total marks: - 48

SECTION 1	/ 16
SECTION 2	/ 16
SECTION 3	/ 16
TOTAL	/ 48

Section I. POLYNOMIALS AND FURTHER GRAPHS (16 marks).

- (a) Show that $x = 1$ and $x = 2$ are roots of the polynomial equation $x^4 - 2x^3 - 7x^2 + 20x - 12 = 0$. (2)
 - (b) Hence sketch the graph of $y = x^4 - 2x^3 - 7x^2 + 20x - 12$ clearly showing any intercepts. (4)
- If $x^3 - 6x^2 + kx - 12 = 0$ has one root equal to the sum of the other two roots, find the value of k . (3)
- Consider the function $f(x) = \frac{x^2}{1-x^2}$.
 - (a) Show that $f(x)$ is even. (1)
 - (b) Find any intercepts of the graph of $y = f(x)$. (1)
 - (c) State the vertical asymptote(s) of the graph of $y = f(x)$. (1)
 - (d) Describe how the function behaves as $x \rightarrow \pm \infty$. (2)
 - (e) Sketch the graph of $y = f(x)$. (2)

Section II. FURTHER TRIGONOMETRY (16 marks). **START A NEW PAGE.**

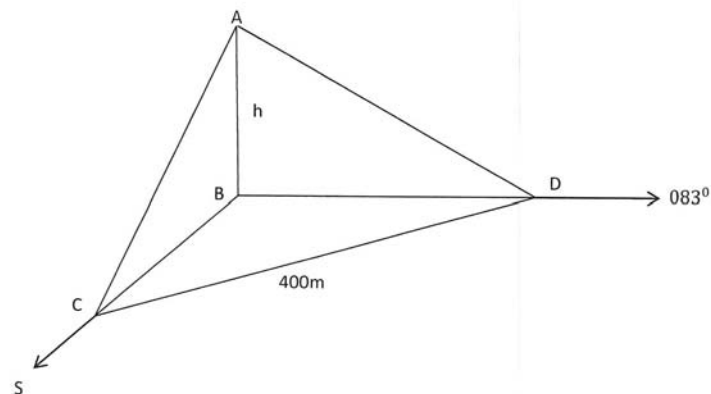
1. Find, in simplest exact form, the value of $\tan 105^\circ$. (3)

2. Prove that $\frac{\sin 2x + \cos x}{\cos 2x + \sin x} = \frac{\cos x}{1 - \sin x}$. (3)

3. Find an expression for $\sec x - \tan x$ in terms of t (where $t = \tan \frac{x}{2}$). (2)

4. Solve $\cos x + \sqrt{3} \sin x = \sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$. (5)

5.



From a point C, due south of the foot of a hill AB, the angle of elevation to the top of the hill is 14° . From a point D, bearing 083° from the foot of the hill, the angle of elevation to the top is 10° . Find the height h of the hill, to the nearest metre, if the distance from C to D is 400 metres. (3)

Section III. PARAMETRIC REPRESENTATION (16 marks). **START A NEW PAGE.**

1. Given the parametric equations $x = 6t, y = 3t^2$.
Eliminate t to find the Cartesian equation of the parabola. (2)

2. The points $P(2ap, ap^2)$ & $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. The chord PQ subtends a right angle at the vertex of the parabola.

(a) Show that $pq + 4 = 0$. (3)

(b) Show that the equation of the chord PQ is $y - \frac{1}{2}(p + q)x + apq = 0$. (3)

3. (a) Show that the equation of the tangent to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$ is $y = px - ap^2$. (3)

(b) The tangent at P meets the axis of the parabola at T . Find the coordinates of the point T . (1)

(c) The normal at P meets the axis of the parabola at U . Show that $TS = US$, where S is the focus of the parabola. (4)

Section I MATHS EXT 1 2012 #1

SOLUTIONS

1 a) Let $P(x) = x^4 - 2x^3 - 7x^2 + 20x - 12$

$$P(1) = (1)^4 - 2(1)^3 - 7(1)^2 + 20(1) - 12 \\ = 1 - 2 - 7 + 20 - 12 \\ = 0$$

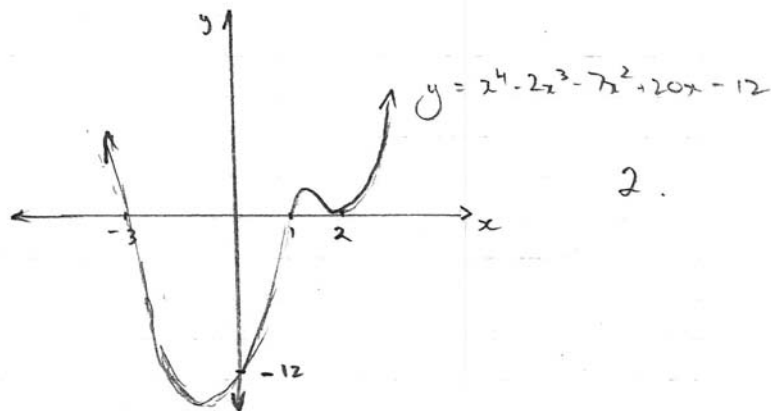
$$P(2) = (2)^4 - 2(2)^3 - 7(2)^2 + 20(2) - 12 \\ = 16 - 16 - 28 + 40 - 12 \\ = 0$$

$\therefore x=1$ & $x=2$ are roots of the eqn.

b) $(x-1)(x-2) = x^2 - 3x + 2$

$$\begin{array}{r} x^2 - 3x + 2 \overline{) x^4 - 2x^3 - 7x^2 + 20x - 12} \\ \underline{x^4 - 3x^3 + 2x^2} \\ x^3 - 9x^2 + 20x \\ \underline{x^3 - 3x^2 + 2x} \\ -6x^2 + 18x - 12 \\ \underline{-6x^2 + 18x - 12} \\ 0 \end{array}$$

$$\therefore P(x) = (x-1)(x-2)(x^2 + x - 6) \\ = (x-1)(x-2)(x+3)(x-2) \\ = (x-2)^2(x-1)(x+3)$$



2. Let the roots be $\alpha, \beta, \alpha + \beta$

$$\alpha + \beta + \alpha + \beta = -\frac{b}{a} \quad \alpha \cdot \beta \cdot (\alpha + \beta) = -\frac{d}{a}$$

$$2\alpha + 2\beta = -\frac{b}{a} \quad \therefore \alpha\beta \cdot 3 = -\frac{d}{a}$$

$$\alpha + \beta = 3$$

$$3\alpha\beta = 12$$

$$\alpha\beta = 4$$

Also $\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = \frac{c}{a}$

$$4 + \alpha \cdot 3 + \beta \cdot 3 = \frac{c}{a}$$

$$4 + 3(\alpha + \beta) = k$$

$$4 + 3(4) = k$$

$$k = 16$$

3. a) $f(x) = \frac{x^2}{1-x^2}$

$$f(-x) = \frac{(-x)^2}{1-(-x)^2}$$

$$= \frac{x^2}{1-x^2}$$

$$= f(x)$$

$\therefore f(x)$ is even

b) If $x=0$, $y = \frac{(0)^2}{1-(0)^2}$

$$= 0$$

$\therefore$ Curve passes thru $(0,0)$

c) $x = \pm 1$

N.B. As $x \rightarrow 1^+$, $y \rightarrow -\infty$

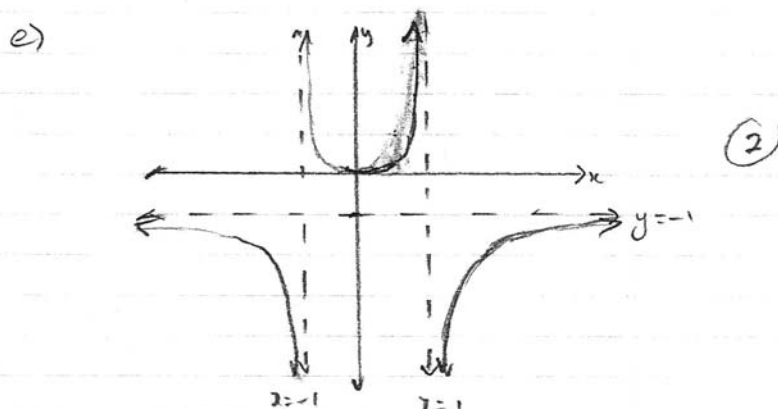
" $x \rightarrow 1^-$, $y \rightarrow \infty$

As $x \rightarrow -1^+$, $y \rightarrow -\infty$

As $x \rightarrow -1^-$, $y \rightarrow \infty$

$$d) \lim_{x \rightarrow \infty} \frac{x^2}{1-x^2} = \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}} \\ = \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x^2} - 1} \quad (2)$$

$\therefore$ As $x \rightarrow \infty$, $\text{den} \rightarrow -1$
As $x \rightarrow -\infty$, $\text{den} \rightarrow -1$



Section II

$$1. \tan 105^\circ = \tan(45^\circ + 60^\circ) \\ = \frac{\tan 45^\circ + \tan 60^\circ}{1 - \tan 45^\circ \tan 60^\circ} \\ = \frac{1 + \sqrt{3}}{1 - 1 \cdot \sqrt{3}} \quad (3) \\ = \frac{1 + \sqrt{3}}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}} \\ = \frac{1 + 2\sqrt{3} + 3}{-2} \\ = -2 - \sqrt{3}$$

$$2. \text{Proof: LHS: } \frac{2 \sin x \cos x - 1 \cos x}{1 - 2 \sin^2 x + \sin x} \\ = \frac{\cos x (2 \sin x + 1)}{1 + \sin x - 2 \sin^2 x} \quad (3) \\ = \frac{\cos x (2 \sin x + 1)}{(1 + 2 \sin x)(1 - \sin x)} \\ = \text{RHS.}$$

$$3. \sec x - \tan x = \frac{1+t^2}{1-t^2} - \frac{2t}{1-t^2} \\ = \frac{1-2t+t^2}{1-t^2} \quad (2) \\ = \frac{(1-t)^2}{(1-t)(1+t)} \\ = \frac{1-t}{1+t}$$

$$4. 1 \cos x + \sqrt{3} \sin x = 2 \cos(x - \alpha) \quad \text{where } \tan \alpha = \frac{\sqrt{3}}{1} \quad (1) \\ \alpha = 60^\circ \quad (1)$$

$$\therefore 2 \cos(x - 60^\circ) = \sqrt{3} \\ \cos(x - 60^\circ) = \frac{\sqrt{3}}{2}$$

$$\therefore x - 60^\circ = -30^\circ, 30^\circ, 330^\circ$$

$$x = 30^\circ, 90^\circ, \cancel{390^\circ}$$

$$\text{Sol}^n \text{ is } x = 30^\circ, 90^\circ.$$

(3)

$$\text{5. In } \triangle ABC, \frac{BC}{h} = \tan 76^\circ$$

$$BC = h \tan 76^\circ$$

$$\text{In } \triangle ABD, BC = h \tan 80^\circ$$

$$\text{In } \triangle BCD, 400^2 = BC^2 + BD^2 - 2BC \cdot BD \cdot \cos 97^\circ$$

$$400^2 = h^2(\tan^2 76^\circ + \tan^2 80^\circ) - 2h^2 \tan 76^\circ \tan 80^\circ \cos 97^\circ$$

$$\therefore 400^2 = h^2 [\tan^2 76^\circ + \tan^2 80^\circ - 2 \tan 76^\circ \tan 80^\circ \cos 97^\circ]$$

$$h^2 = \frac{400^2}{\tan^2 76^\circ + \tan^2 80^\circ - 2 \tan 76^\circ \tan 80^\circ \cos 97^\circ}$$

$$h = 55 \text{ metres (nearest metre)}$$

(3)

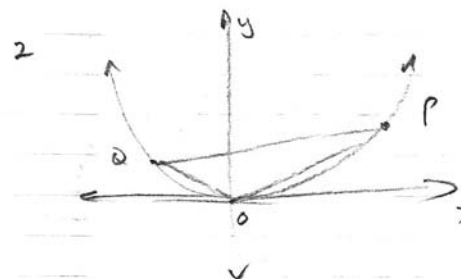
Section III

$$1. x = 6t \Rightarrow t = \frac{x}{6}$$

$$\therefore \text{If } y = 3t^2 \\ y = 3\left(\frac{x}{6}\right)^2 \\ y = \frac{3x^2}{36}$$

$$y = \frac{x^2}{12} \quad \text{or} \quad x^2 = 12y.$$

(2)



$$\text{a) M of PQ} = \frac{ap^2 - 0}{2ap - 0} \\ = \frac{p}{2}$$

$$\text{M of QO} = \frac{aq^2 - 0}{2aq - 0} \\ = \frac{q}{2}$$

(3)

$$\therefore \frac{p}{2} \times \frac{q}{2} = -1$$

$$\text{ie } \frac{pq}{4} = -1$$

$$pq = -4$$

$$\therefore pq + 4 = 0$$

$$\begin{aligned}
 \text{b) M of } PO &= \frac{ap^2 - aq^2}{2ap - 2aq} \\
 &= \frac{a(p-q)(p+q)}{2a(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}$$

∴ Eqn of PO is

$$y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$$

$$y - ap^2 = \frac{1}{2}(p+q)x - \frac{1}{2}(p+q)2ap$$

$$y - ap^2 = \frac{1}{2}(p+q)x - ap^2 - apq$$

$$\text{i.e. } y - \frac{1}{2}(p+q)x + apq = 0$$

$$\begin{aligned}
 \text{3 a) If } x^2 &= 4ay \\
 y &= \frac{x^2}{4a} \\
 y' &= \frac{2x}{4a} \\
 &= \frac{x}{2a}
 \end{aligned}$$

$$\text{At } (2ap, ap^2) \quad m = \frac{2ap}{2a}$$

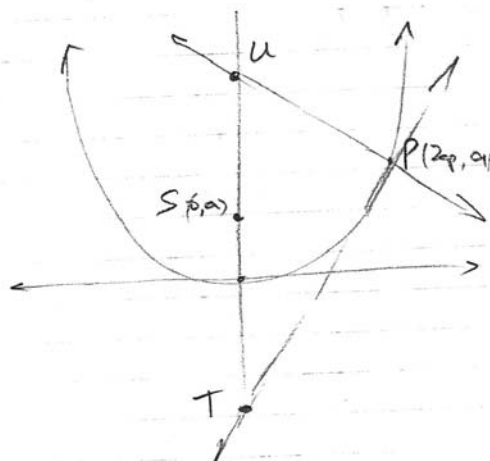
∴ the gradient of the tangent = p

∴ the eqn of the tangent is

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$



b) Axis of parabola is $x=0$

$$\text{When } x=0, y = p(0) - ap^2$$

$$y = -ap^2 \quad (1)$$

$$\therefore U \text{ is } (0, -ap^2)$$

c) the gradient of the normal is $-\frac{1}{p}$

∴ the eqn of the normal is

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = ap^3 + 2ap$$

$$\text{When } x=0, py = ap^3 + 2ap$$

$$y = ap^2 + 2a$$

$$\therefore U \text{ is } (0, ap^2 + 2a)$$

$$S \text{ is } (0, a)$$

$$\therefore TS = a + ap^2$$

$$US = ap^2 + 2a - a$$

$$= a + ap^2$$

$$\therefore TS = US$$